

ESSAY: Introduction to Computer Vision

CODE: 503080

I. Rules

- Each essay is conducted by a group of **one to three students**.
- The essay consists of 2 parts: the Programming part and the Report part.
- Conduct the essay with a topic in the following link:
<https://docs.google.com/spreadsheets/d/1JeZa3lakFdD08HSrazqioevNz-yDXdbvHDZjVlIcSyw/edit?usp=sharing>

II. Programming part (3.0 points)

Do a demo for your assigned topic:

- Select a public well-known dataset: split train/test data (if not exist)
- Code your model
- Evaluate your model's performance using the following metrics: Precision, Recall, F1-score, mAP@0.5, and mAP@0.5:0.95

III. Report part (7.0 points)

1. The report must be submitted in **PDF format**, and the content must be written based on the report/essay format of the Faculty of Information Technology. **In case students do not follow the Faculty's format, they will receive 0 points for the Report part.**

2. The report must include the following contents:

- a. Chapter 1: <Your topic name> **(5.0 points)**

Write a detailed content of your assigned topic:

- Purpose of the model/algorithm & practical applications – *1 point*
- Detailed architecture of the model/algorithm (with clear diagrams or drawings) – *2 points*
- Detailed explanation of the following metrics: Precision, Recall, F1-score, mAP@0.5, and mAP@0.5:0.95 – *2 points*

- b. Chapter 2: Programming results **(2.0 points)**

- Detailed explanation about the dataset: data source, number of images in the train/test data, class labels distribution – *0.75 point*
- Insert your results (the model's performance using various metrics) in the “Programming part” – *1.25 points*
- Insert your Google Drive link containing the original source code if you conduct the Programming part in a notebook

The images must be clear, and properly laid out. The images captions and descriptions are also required.

IV. Submission guideline

- Filenames of the source code and the report files must be the **Student IDs**, for ex.,
 - A group of only one student with student ID 521H1495 will submit a Python source file named **521H1495.py** and a report file named **521H1495.pdf**
 - A group of two students with student IDs 521H1234 and 522H4321 will submit a Python source file named **521H1234_522H4321.py** and a report file named **521H1234_522H4321.pdf**
- Students submit **Python source files (file extension is .py)**, and a **report file** to the **"MidTerm_Essay"** assignment on Elearning website of the practical class.
- Students must ensure that the Python source files are not corrupted during execution. The source code with errors will not be scored.
- Python source files must be saved in the correct format (**file extension is .py**). The source files in the wrong format will not be scored.

V. Regulations

- The result of this essay will be the Midterm score.
- **Student who copy their friends's essay will be scored 0.**
- **If a student's work shows signs of copying each other, the student will attend an interview with the lecturer.**

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